## Weekly Lesson Plan – Week 1 · Semester 1 · 2026–27 · Da Vinci School

**Teacher: Gavaldon | Week of: Aug 10, 2026 | Course: Manufacturing Engineering Technology I**

**TEKS / ELPS:**

TEKS: §127.15(d)(1) CTE employability skills - safety, workplace expectations & teamwork; §127.813(d)(5) mechanical & fluid systems; (d)(6) electrical & thermal systems (systems overview); §127.813(d)(1)(A) use CAD to complete a design; §127.15(d)(1) problem-solving & teamwork; §127.813(d)(1)(A) complete a design; (d)(1)(B) analyze a design in a modeling environment; §127.15(d)(1) employability skills - communicate a design solution (presentation)
ELPS: c.1A, c.2D, c.3C, c.4G, c.5B

**Aim of the Lesson / Objective:** *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

| MONDAY | TUESDAY | WEDNESDAY | THURSDAY | FRIDAY |
|---|---|---|---|---|
| Students will classify manufacturing-lab hazards, apply the NIOSH hierarchy of controls to select the most effective control for a given task, complete a Job Safety Analysis (JSA) with a likelihood x severity risk score, and set up a dated, numbered engineering design notebook as a professional record. | Students will model production as an input -> process -> output system with a control loop, identify the six inputs for a real product, classify job/batch/repetitive/continuous production against the volume-variety product-process matrix, and compute a value-added ratio and per-unit cost to justify a production-type choice. | Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan). | Students will finish their final design, write a structured AI prompt, and generate + submit a product image. | Students will present their product - communicating the problem it solves, their final design, and how it works - and give supportive peer feedback. |

**Employed Language Domain(s) Objective:** *How students achieve the objective – Listening, Speaking, Reading, Writing. Students will…*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.

**Assessment of Content Objective:** *Instrument used to determine mastery. Ex: Exit Ticket.*

Students will be assessed by written, group, or oral responses throughout the class period.

**Set (Bell Work):** *A motivating action that engages students in the objective – question, fact, activity. Essential Question:*

| MONDAY | TUESDAY | WEDNESDAY | THURSDAY | FRIDAY |
|---|---|---|---|---|
| Look at the shop photo. List every hazard you can spot. | How is making 10,000 cars different from making one wedding cake? | What is the purpose of the engineering design cycle? | Watch: how a product improves, version to version | Write your script before you present |

**Methodology / Steps:** *What it takes to teach the objective so 100% can pass the assessment. Teacher will… Teams will… Individuals will…*

| MONDAY | TUESDAY | WEDNESDAY | THURSDAY | FRIDAY |
|---|---|---|---|---|
| Step 1: Bell ringer / essential question.<br>Step 2: Bell-work: list every hazard you can spot in a shop photo (5 min)<br>Step 3: Why safety is a professional standard + the six hazard classes (10 min)<br>Step 4: Hierarchy of controls, machine guarding, LOTO & PPE (12 min)<br>Step 5: Worked Job Safety Analysis + guided you-try (10 min)<br>Step 6: The engineering notebook as a legal/professional record — set up Year-2 sections (5 min)<br>Step 7: Exit ticket + sign safety agreement (3 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell-work: how is making a car different from a wedding cake? (5 min)<br>Step 3: The systems model + the six inputs + process families (8 min)<br>Step 4: Value-added vs the 7 wastes + worked VA-ratio (7 min)<br>Step 5: Toyota Production System video + active-viewing notes (8 min)<br>Step 6: Production types, product-process matrix & economies of scale (7 min)<br>Step 7: Guided practice: classify & justify + start worksheet (7 min)<br>Step 8: Exit ticket (3 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell-work: purpose of the engineering design cycle (5 min)<br>Step 3: Present: the 5-step engineering design process (16 min)<br>Step 4: Pick a real problem (list 3, choose 1) (7 min)<br>Step 5: Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials) (17 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell-work + video (start 1:28): name 2 improvements and why each matters (8 min)<br>Step 3: Model the notebook page (prompt on top, sketch on bottom) + copy the prompt (10 min)<br>Step 4: Finish final design, then generate the product image in Gemini (20 min)<br>Step 5: Submit the image to Google Classroom (7 min) | Step 1: Bell ringer / essential question.<br>Step 2: Bell-work: write your 4-sentence presentation script (5 min)<br>Step 3: Model: the four things every presenter says + how to be a great audience (8 min)<br>Step 4: Presentations: each student shows their AI product image + notebook page (~1 min each) (32 min)<br>Step 5: Exit ticket: one thing you would design next (5 min) |

**Resources:** *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.

**Re-Teaching Activities:** *What you prepare if students do not master the objective.*

Review during bellwork; small-group re-teach; peer support.

**Reflection:** *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state what they accomplished and what worked best.

**Study Skills:** *Skills implemented by students – note-taking, summarizing, teamwork, etc.*

Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.

**SPED & 504 Accommodations / Modifications:** *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

| **ELL Accommodations:** *Codes.* |
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| Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). |